

🔥 XAI: Understanding Course Foundations and Motivations

FAMAF - UNC

September 9, 2025



**eXplainable
Artificial Intelligence**

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Agenda ---

- Course Goals & Logistics
- Model Understanding: Use Cases
- Inherently Interpretable Models vs. Post-hoc Explanations
- Defining and Understanding Interpretability

🍷 Goals of this Course ---

Learn and improve upon the state-of-the-art literature on ML interpretability and explainability:

- Understand where, when, and why is interpretability/explainability needed
- Read, present, and discuss research papers
- Formulate, optimize, and evaluate algorithms
- Implement state-of-the-art algorithms; Do research!
- Understand, critique, and redefine literature

EMERGING FIELD!!

Course Overview ---

- ➊ **Foundations and Human Factors** - definitions, taxonomies, and cognitive aspects
- ➋ **Interpretability and Explainability Methods** - inherently interpretable models and post-hoc explanations
- ➌ **Advanced Techniques and Evaluation** - attention-based explanations and quality metrics
- ➍ **Interpretability of Large Models** - mechanistic interpretability and LLM understanding
- ➎ **Emerging Frontiers of XAI** - automated circuit discovery and multimodal techniques

🍷 Who should take this course? ---

- Course particularly tailored to students interested in **research** 📝 on interpretability/explainability
 - ▶ **Not a surface level course!** 🔥
 - ▶ **Not just applications!** 🔥
- Goal is to push you to question existing work and make new contributions to the field

Class Format ---

- Course comprises of lectures, guests, and student presentations
- Each lecture will cover: at least 2 papers  
- Students are **expected** to “at least” skim through the papers beforehand 
- Students will divide into groups

Each breakout group is expected to come up with:

- A list of 2 to 3 weaknesses of each of the works discussed
- Strategies for addressing those weaknesses

Research Projects - AKA Final Exam ---

Requirements

- Short research paper (max 4 pages)
- Teams of 2 students
- Target: Local conference submission

Assessment

- Ongoing feedback from entire course
- Course approved when paper is submitted

Background ---

Understanding:

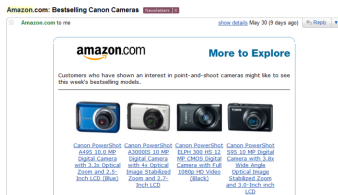
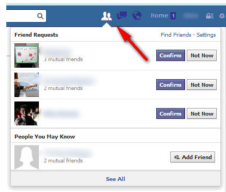
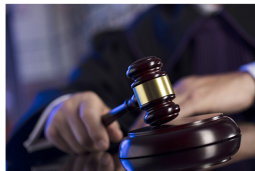
- Linear algebra
- Probability
- Algorithms
- Machine learning
- Programming in Python, Numpy, Sklearn

Familiarity with:

- Statistics
- Optimization

🍷 Motivation ---

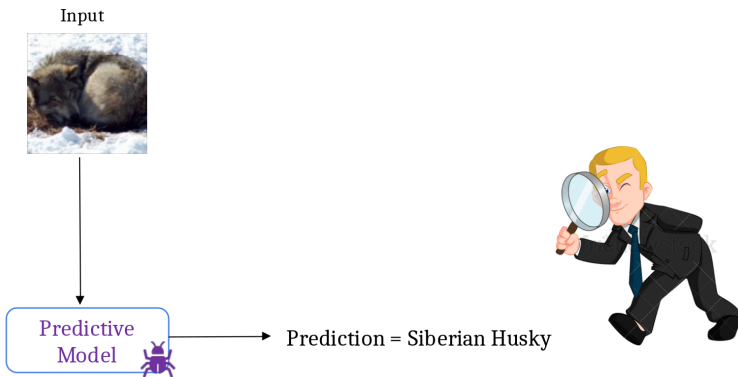
Machine Learning is EVERYWHERE!!



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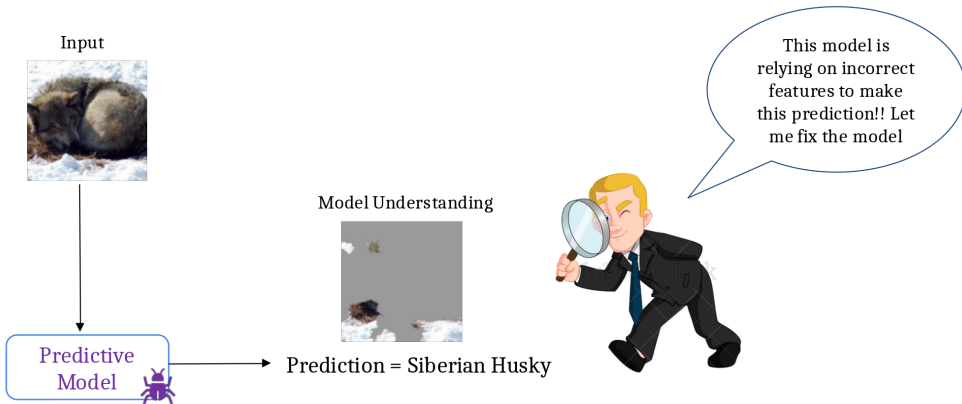
🍷 Motivation: Why Model Understanding? ---

Example: Image classification model



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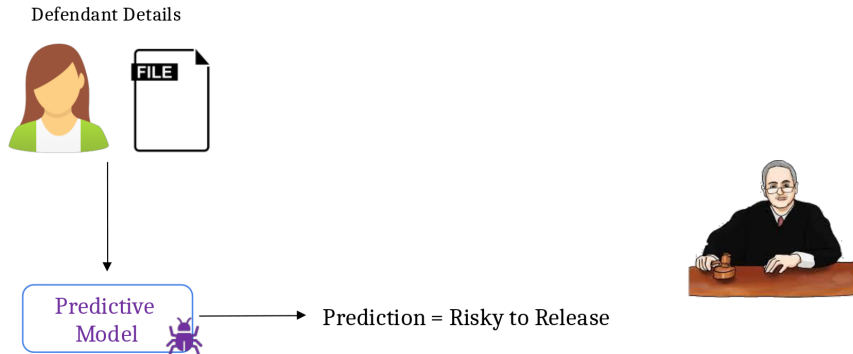
Example: Image classification model



Model understanding facilitates debugging

🍷 Motivation: Why Model Understanding? ---

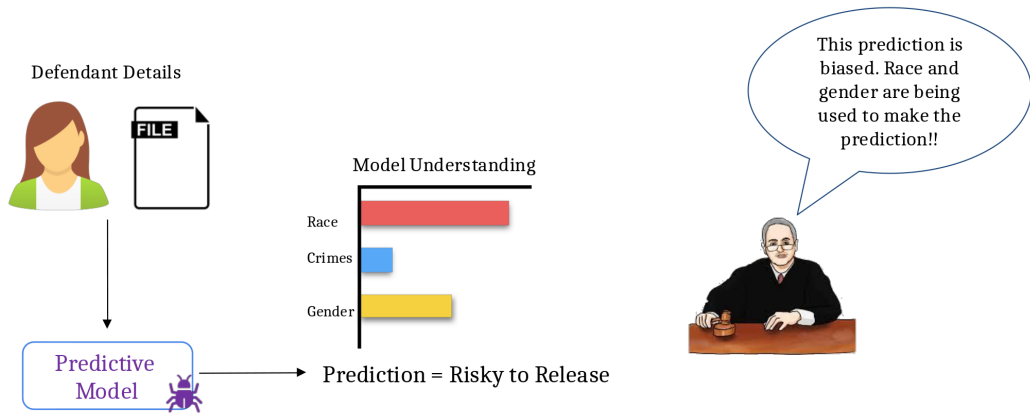
Example: Criminal justice risk assessment



(???)

🍷 Motivation: Why Model Understanding? ---

Example: Criminal justice risk assessment



Model understanding facilitates bias detection

(???)

Motivation: Why Model Understanding? ---

Example: Loan application system

Loan Applicant Details



Predictive
Model



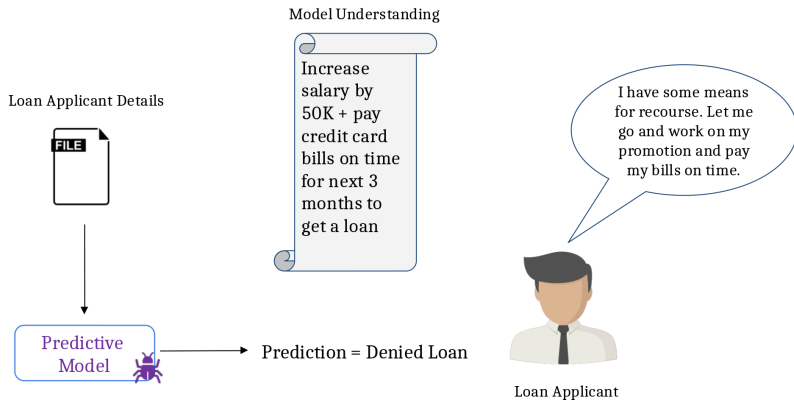
Prediction = Denied Loan



Loan Applicant

🍷 Motivation: Why Model Understanding? ---

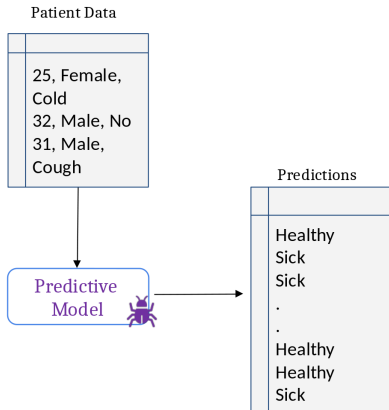
Example: Loan application system



Model understanding helps provide recourse to individuals who are adversely affected by model predictions

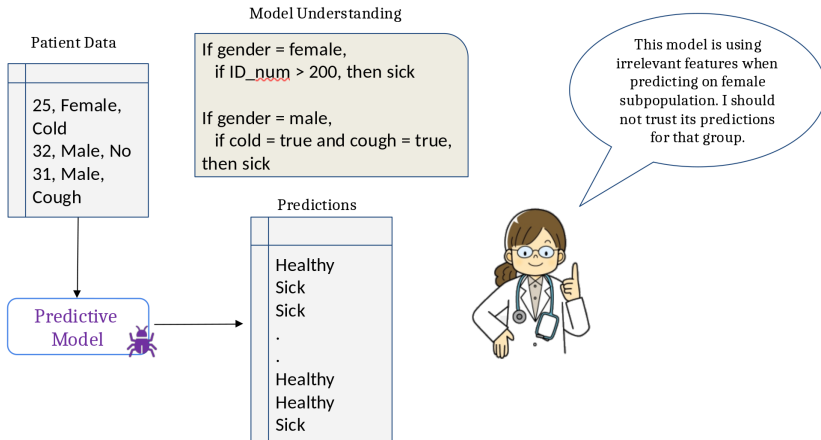
🍷 Motivation: Why Model Understanding? ---

Example: Medical diagnosis system



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Example: Medical diagnosis system



Model understanding helps assess when to trust predictions

Motivation: Why Model Understanding? ---

Summary of Use Cases

Utility

Debugging

Bias Detection

Recourse

If and when to trust model predictions

Vet models to assess suitability for deployment

Motivation: Why Model Understanding? ---

Summary of Use Cases

Utility

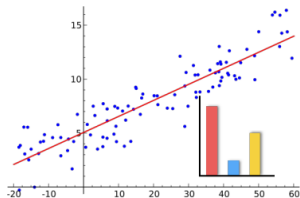
- Debugging
- Bias Detection
- Recourse
- If and when to trust model predictions
- Vet models to assess suitability for deployment

Stakeholders

- End users (e.g., loan applicants)
- Decision makers (e.g., doctors, judges)
- Regulatory agencies (e.g., FDA, European commission)
- Researchers and engineers

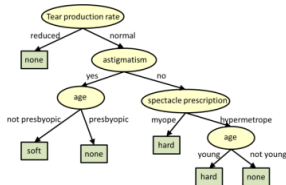
🍷 Achieving Model Understanding ---

Take 1: Build inherently interpretable predictive models



Linear regression

(???, ??)



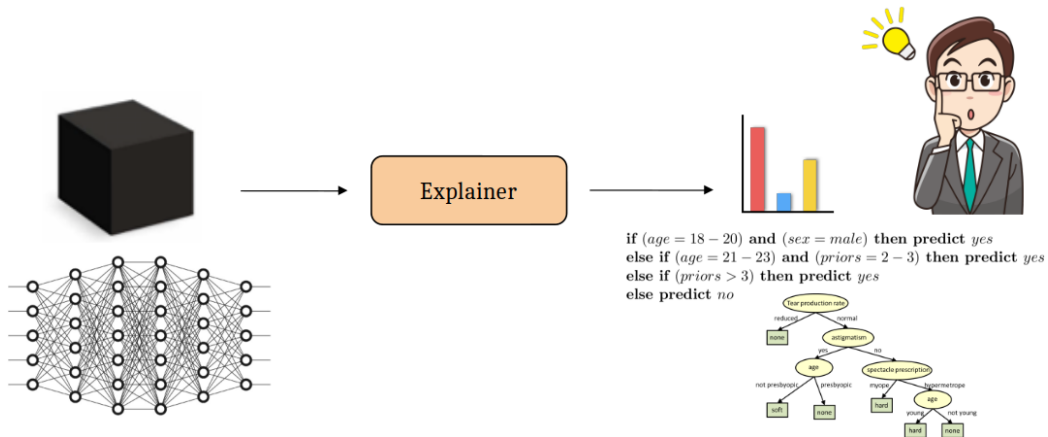
Decision trees

if (*age* = 18 – 20) and (*sex* = *male*) then predict *yes*
else if (*age* = 21 – 23) and (*priors* = 2 – 3) then predict *yes*
else if (*priors* > 3) then predict *yes*
else predict *no*

Rule-based models

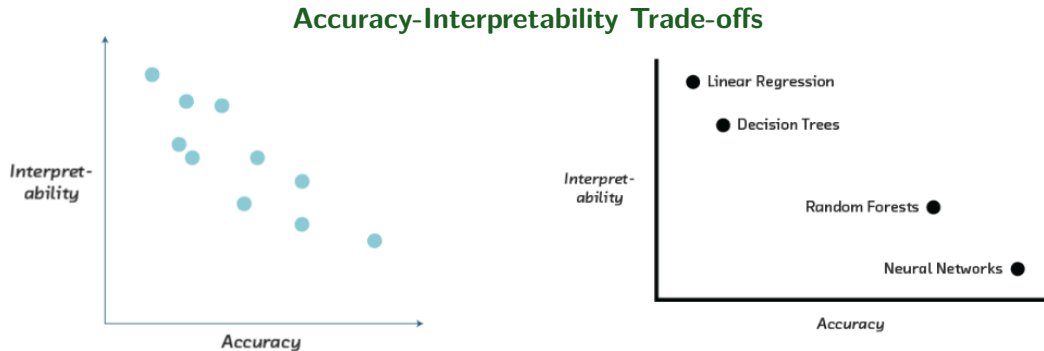
🧠 Achieving Model Understanding ---

Take 2: Explain pre-built models in a post-hoc manner



(???, ???, ???)

🍷 Inherently Interpretable Models vs. Post hoc Explanations ---

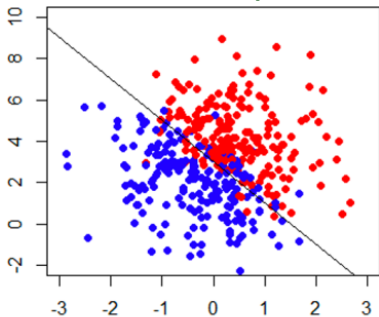


In certain settings, accuracy-interpretability trade offs may exist.

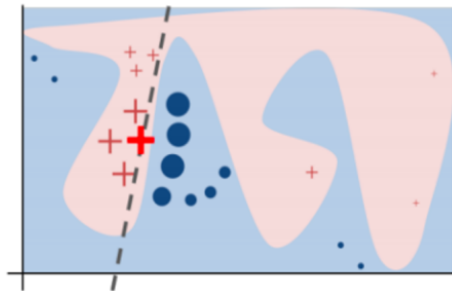
(Cireřan et. al. 2012, Caruana et. al. 2006, Frosst et. al. 2017, Stewart 2020)

🍷 Inherently Interpretable Models vs. Post hoc Explanations ---

Example scenarios: Simple vs complex boundaries



can build interpretable + accurate models



complex models might achieve higher accuracy

In certain settings, accuracy-interpretability trade offs may exist.

🍷 Inherently Interpretable Models vs. Post hoc Explanations ---

Sometimes, you don't have enough data to build your model from scratch. And, all you have is a (proprietary) black box!

(???)

🍲 Inherently Interpretable Models vs. Post hoc Explanations ---

Recommendation

If you can build an interpretable model which is also adequately accurate for your setting,

DO IT!

*Otherwise, **post hoc explanations** come to the rescue!*

🍷 Inherently Interpretable Models vs. Post hoc Explanations ---

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Let's get into some details!

Next Up! ---

- Define and evaluate interpretability somewhat! 🧠
- Taxonomy of interpretability evaluation
- Taxonomy of interpretability based on applications/tasks
- Taxonomy of interpretability based on methods

🍷 Defining and Understanding Interpretability: Motivation for Interpretability ---

- ML systems are being deployed in complex **high-stakes settings**
- Accuracy alone is no longer enough
- **Auxiliary criteria are important:**
 - ▶ Safety
 - ▶ Nondiscrimination
 - ▶ Right to explanation

🍷 Motivation for Interpretability (cont.) ---

- Auxiliary criteria are often **hard to quantify** (completely):
 - ▶ E.g.: Impossible to enumerate all scenarios violating safety of an autonomous car

Fallback option: *Interpretability*

If the system can explain its reasoning, we can verify if that reasoning is sound w.r.t. auxiliary criteria

🍲 Prior Work: Defining and Measuring Interpretability ---

Bad News

Little consensus on what interpretability is and how to evaluate it.

Interpretability evaluation typically falls into:

① Evaluate in the context of an application

- ▶ If a system is useful in a practical application or a simplified version, it must be interpretable

② Evaluate via a quantifiable proxy

- ▶ Claim some model class is interpretable and present algorithms to optimize within that class
- ▶ E.g. rule lists

"You will know it when you see it!"

🍷 Lack of Rigor? ---

Yes and No

Previous notions are reasonable

- **However:**

- ▶ Are all models in all “interpretable” model classes equally interpretable?
 - ★ Model sparsity allows for comparison
- ▶ How to compare a linear model with a decision tree?
- ▶ Do all applications have same interpretability needs?

Important to formalize these notions!!!

🍷 What is Interpretability? ---

Definition

Ability to explain or to present in understandable terms to a human

No clear answers in psychology to:

- What constitutes an explanation?
- What makes some explanations better than the others?
- When are explanations sought?

🍷 When and Why Interpretability? ---

Not all ML systems require interpretability

- E.g., ad servers, postal code sorting
- No human intervention
- **No explanation needed because:**
 - ▶ No consequences for unacceptable results
 - ▶ Problem is well studied and validated well in real-world applications → trust system's decision

When do we need explanation then?

🍷 When and Why Interpretability? ---

- **Incompleteness in problem formalization**
 - ▶ Hinders optimization and evaluation
- **Incompleteness $\neq$ Uncertainty**
 - ▶ Uncertainty can be quantified
 - ▶ E.g., trying to learn from a small dataset (uncertainty)

🍷 Incompleteness: Illustrative Examples ---

Scientific Knowledge

- E.g., understanding the characteristics of a large dataset
- Goal is abstract

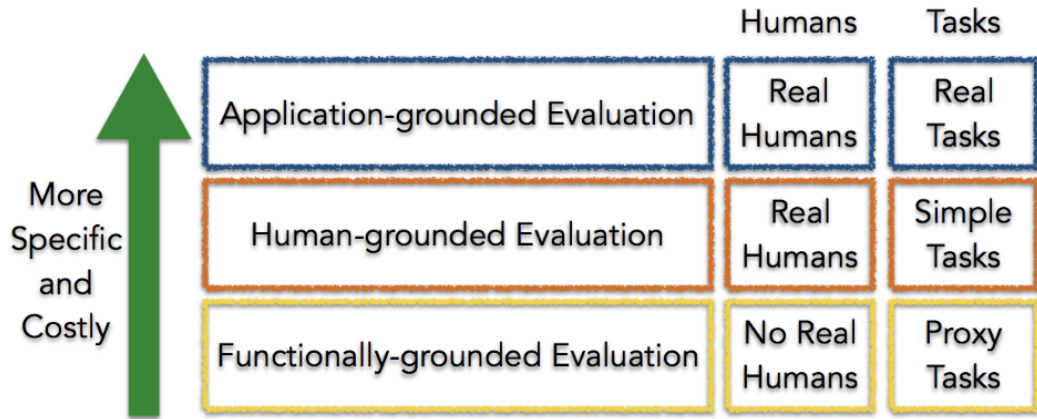
Safety

- End to end system is never completely testable
- Not possible to check all possible inputs

Ethics

- Guard against certain kinds of discrimination which are too abstract to be encoded
- No idea about the nature of discrimination beforehand

🍷 Taxonomy of Interpretability Evaluation ---



Claim of the research should match the type of the evaluation!

🍷 Application-grounded evaluation ---

- Real humans (domain experts), real tasks
- Domain experts experiment with **exact application task**
- Domain experts experiment with a **simpler or partial task**
 - ▶ Shorten experiment time
 - ▶ Increases number of potential subjects
- Typical in *Human-Computer Interaction* (HCI) and visualization communities

🍷 Human-grounded evaluation ---

Real humans, simplified tasks

- Can be completed with lay humans
- Larger pool, less expensive

Potential experiments:

- Pairwise comparisons
- Simulate the model output
- What changes should be made to input to change the output?

🍷 Functionally-grounded evaluation ---

No humans, just proxies

- Appropriate for a class of models already validated (E.g., decision trees)
- A method is not yet mature
- Human subject experiments are unethical
- What proxies to use?

Potential experiments:

- Complexity (of a decision tree) compared to other models of the same (similar) class
 - ▶ How many levels? How many rules?

🍲 Open Problems: Design Issues ---

- What proxies are best for what real world applications? 🙌
- What factors to consider when designing simpler tasks in place of real world tasks? 🙌

🍲 Taxonomy based on applications/tasks ---

Global vs. Local

- High level patterns vs. specific decisions

Degree of Incompleteness

- What part of the problem is incomplete? How incomplete is it?
- Incomplete inputs or constraints or costs?

Time Constraints

- How much time can the user spend to understand explanation?

🍷 Taxonomy based on applications/tasks (cont.) ---

Nature of User Expertise

- How experienced is end user?
- Experience affects how users process information
- E.g., domain experts can handle detailed, complex explanations compared to opaque, smaller ones

Note:

These taxonomies are constructed based on intuition and are not data or evidence driven. They must be treated as hypotheses.

Taxonomy based on methods ---

Basic units of explanation:

- Raw features? E.g., pixel values
- Semantically meaningful? E.g., objects in an image
- Prototypes?

Number of basic units of explanation:

- How many does the explanation contain?
- How do various types of basic units interact?
- E.g., prototype vs. feature

Taxonomy based on methods (cont.) ---

Level of compositionality:

- Are the basic units organized in a structured way?
- How do the basic units compose to form higher order units?

Interactions between basic units:

- Combined in linear or non-linear ways?
- Are some combinations easier to understand?

Uncertainty:

- What kind of uncertainty is captured by the methods?
- How easy is it for humans to process uncertainty?

🍷 Relevant Conferences to Explore ---

ML/AI Venues:

- ICML, NeurIPS, ICLR
- UAI, AISTATS, KDD, AAAI

Ethics/HCI Venues:

- FAccT, AIES
- CHI, CSCW, HCOMP

Closing ---

- Say hi to your neighbors! Introduce yourselves!
- What topics are you most excited about learning as part of this course?
- Are you convinced that model interpretability/explainability is important?
- Do you think we can really interpret/explain models (correctly)?
- What is your take on inherently interpretable models vs. post hoc explanations? Would you favor one over the other? Why?

References --- |